



SOP Title:

SP3 Protocol for dental calculus

SINGLE-POT, SOLID-PHASE ENHANCED, SAMPLE-PREPARATION

Document Number: #Proteomics102

SOP Originator: Palaeoproteomics PhD	Approving Position: Palaeoproteomics Lab Head	Effective Date: 01/Jan/2019
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LOCATION:

Palaeoproteomics Clean Rooms

PRINCIPAL:

Extraction and digestion of proteins from archaeological dental calculus for analysis by LC-MS/MS.

IF YOU USE THIS FOR PUBLICATION PLEASE CITE:

Cleland, Timothy P. 2018. "Human Bone Paleoproteomics Utilizing the Single-Pot, Solid-Phase-Enhanced Sample Preparation Method to Maximize Detected Proteins and Reduce Humics." *Journal of Proteome Research* 17(11): 3976–83.

Hughes, C. S., Moggridge, S., Müller, T., Sorensen, P. H., Morin, G. B. & Krijgsveld, J. 2019. "Single-pot, solid-phase-enhanced sample preparation for proteomics experiments". *Nat. Protoc.* 14, 68–85.

COSHH RE

Risk assessment is evaluated by the Lab Manager/Safety Officer. The safety data sheets pertaining to the lab are in an extra folder stored in the lab and all details about hazardous goods or other chemical components can be found in the database: <https://www.dguv.de/ifa/gestis/gestis-stoffdatenbank/index.jsp>.

TRAINING REQUIRED

Clean Lab Procedures, Pipetting, Centrifuge, Balance, Proteomics wet chemistry training

Please see *Kristine Richter* and *Jana Zech* or *Sandra Hebestreit* (Technicians) prior to starting any work detailed in this SOP if you have not completed all of the training listed above.



I SECTION: GENERAL CLEAN ROOM INSTRUCTIONS

Cleaning Reagents

- Bleach for surface cleaning: filled in red squeeze bottles (5% bleach cleaning solution); stock: large blue bottles “Klorix” under the sink for refill
- Bleach for tool cleaning in yellow topped bottles (half 10-15% reagent grade bleach stock solution found in the refrigerator in the Isotope lab and half ultrapure water)
- Water in blue bottles (100% ultrapure water)
- 70% Ethanol in orange or green topped bottles: fill the bottle to approximately 70% with 100% EtOH found in the flammables cabinet and then fill up with MilliQ water)

Cleaning of surfaces and tools

- Make sure to clean inside the hood with bleach, water, and then Ethanol before and after working.
- If you see any spills, clean them up with water first before you use bleach. If there is white salt residue on a surface after cleaning, you can clean it with ultrapure water.
- For cleaning tools use bleach first, then ultrapure water, and then Ethanol.
- NEVER use bleach to clean up Guanidine hydrochloride as it will create chlorine gas, which is extremely dangerous.

Lab Principles and Chemical Handling

- You should never pipette out of a stock solution bottle. Aliquot by pouring the amount of stock solution you need into a tube of a suitable volume (5 ml, 15 ml, or 50 ml) in the reagent preparation hood for your working solutions.
- Remember to label all of your working solutions with the name of the chemical and dispose of them properly when you are done with the protocol.
- Please make sure that everything stored in the refrigerator or freezer is in a labelled box or bag. The label should include your name or initials and the date.



II SECTION: REAGENT INSTRUCTIONS SHEET

General Instructions

Used glassware for preparing must be combusted before! A stock of combusted glassware is prepared and placed in the DA Proteomics Lab.

Solutions should be prepared under the specific dead air hood for reagent preparation.

TFA handling only under the fume hood in the Metabolite Lab (or Isotope Lab).

Prepare the Guanidine Hydrochloride solution last!

Every bottle and tubes containing stock solutions shall be labelled with the name of the chemical, concentration, date and your initials.

Bead Solution (20 µg/µl; Shelf Life: 1 month at 4°C, mix before use)

Beads:

- SeraMag SpeedBead Hydrophilic 15 ml, GE Healthcare, GE45152105050250 (stock: 50 mg/ml)
- SeraMag SpeedBead Hydrophobic 15 ml, GE Healthcare, GE65152105050250 (stock: 50 mg/ml)

Instructions:

- Invert the bottles of bead stock solutions and knock(!) them on the counter to gain an equal distribution of the beads in the stock solution (this step is very important, beads tend to be very sticky!). Make sure you can visibly see that the beads are no longer stuck to the walls of the bottle before pipetting.
- Pipette 200 µl of the hydrophobic and hydrophilic each into one 1.5 ml tube (combine them)
- Place the tube on the magnetic rack, remove the stock supernatant and discard
- Replace with 1 ml molecular grade water and mix
- Repeat the wash 2x
- After the final wash, add 1000 µl molecular grade water to the beads and resuspend to mix (working concentration is now 20 µg/µl)

} First wash

Aliquots for Reduction and Alkylation solution (Shelf Life: 6 months at 4°C – solution made during extraction)

Reduction and alkylation lysis solution is 1 ml of 100 mM CAA (2-Chloroacetamide), 100mM TCEP (Tris(2-carboxyethyl)phosphine) prepared during extraction. To facilitate this, weigh out:

- 28.66 mg of TCEP into a 1.5 or 2 ml tube.
- 9.35 mg of CAA into a 1.5 ml black tube as it is slightly light sensitive



6 M Guanidine Hydrochloride (GuHCl) (Shelf Life: 3 months at 4°C)

*****EXTREMELY IMPORTANT!***** Never mix or clean up GuHCl with bleach. This will create chlorine gas which is highly toxic and would necessitate the immediate evacuation of our lab. Notify to the lab managers immediately.

- Weigh out the 28.66 g GuHCl into a 50 ml tube
- Aliquot out around 40 ml of water into a 50 ml tube
- Now mix the GuHCl with 25 ml molecular grade water and check the volume.
- Fill up to 50 ml with molecular grade water and transfer the solution to a 50 ml glass bottle for long-term storage.

Note: Prepare the GuHCl solution last. After preparing, wipe down all surfaces with water first before cleaning the surfaces with bleach.

100% and 80% Ethanol (Shelf Life: 1 month at 4°C)

100%: Aliquot out 50 ml of HPLC grade Ethanol into a 50 ml bottle

80%: Add 40 ml of HPLC grade Ethanol into a 50 ml bottle. Add 10 ml of molecular grade water.

100 mM Ammonium bicarbonate (AmBic, pH 8, Shelf life: 6 months at 4 °C)

- Weight out 395 mg of AmBic powder.
- Dissolve it in 50 ml molecular grade water and mix through gently shaking.

5% Trifluoroacetic acid (TFA) (Shelf Life: 3 months at room temperature)

Note: Trifluoroacetic acid (TFA) is very hazardous (corrosive)! Handling the stock TFA should only be performed under a fume hood. Always pipette water first, then slowly add the acid! Glass pipettes or syringes, not plastic, should be used to pipette fully concentrated TFA to avoid plastic contamination. Working solutions with TFA for this protocol are 5% and can be handled outside of a fume hood and with plastic pipettes.

- For 20 ml, dilute 1 ml TFA stock solution using the special glass syringe (not plastic) in 19 ml ultrapure water and mix by shaking carefully.

100% MeOH (Shelf life: 1 month at 4 °C)

Aliquot out into a 50 ml bottle



3% Acetonitrile (ACN), 0.1% Trifluoroacetate (TFA) (Shelf life: 3 Months at 4 °C)

- First pour roughly 80 ml molecular grade water into a 100 ml glass bottle
- Add 3 ml ACN by pipetting and fill up to 100 ml with molecular grade water
- Add 100 µl TFA with a glass syringe (Metabolite Lab, fume hood)

60% Acetonitrile (ACN), 0.1% TFA (Shelf life: 3 Months at 4 °C)

- First pour 20 ml molecular grade water in a 50 ml glass bottle
- Add 30 ml ACN
- Add 50 µl TFA with a glass syringe (Metabolite Lab, fume hood)

Trypsin solution (Stable once resuspended for 1 month at -20°C)

Freeze dried Trypsin (20 µg, stored at -20 °C) is resuspended in 100 µl of Trypsin resuspension buffer (pink cap). The final concentration is now 0.2 µg/µl. Transfer the trypsin solution into a labelled 0.5 ml tube (content, date, name) and store at -20°C.



III SECTION: SAMPLE PREPARATION

This protocol is for moderate to well-preserved calculus samples. Normal batches consist of 2 to 10 calculus samples, 1 positive control from an archaeological sheep bone, and 1 extraction blank. The samples are prepared and demineralized for 4-7 days before a two day SP3 extraction protocol.

Note: All steps should take place in the sample preparation hood.

Before start to work:

- Wipe down all surfaces
- Remove the number of samples tubes you will need (storage cabinet)
- Aliquot out EDTA from the stock solution bottle (reagents cabinet)
- Retrieve the sheep bone positive control from the freezer (changing room freezer)

1. Weighing out the sample

- Weigh out 5-10 mg calculus per sample in a 2 ml Eppendorf tube and record the weight in your lab notebook. Label the side of the tube with the DA proteomics lab number (refer to Google Sheet to label your DA protein lab number), the date, and your initials. Make sure the top is labelled with the DA proteomics lab number and your initials. Make sure to note the DA ARCHive code for each sample in your lab notebook and in the Google Sheet.
- How to label tubes:
 - Top: DA 206 (this is the Protein lab code from the Google Sheet)
SW (Initials)
 - Side: DA206
2019 Apr 11 (Date in this format)
SW (Initials)
- Weigh out 2-3 mg of the sheep bone positive control into a 1.5 ml/ 2 ml tube or use a prepared aliquot (both stored in the -20 freezer in the changing area of the proteomics lab). Label with "Positive Control", the date, and your initials.
- Label an empty 2 ml Eppendorf tube with the DA code for your extraction blank. Label with "Ex. Blank", the date, and your initials on the side of the tube.

Note: Every step performed with the calculus samples should be recreated with both the positive control and extraction blank.



2. Demineralization

- Add 500 µl of 0.5 M EDTA (from your aliquot) to each sample and place on the rotator.
- Samples (including the control and blank) should be left on the rotator for 4-7 days (until the calculus and bone samples look visibly demineralized. Your blank should remain clear).

Note: The demineralized calculus can range from being completely invisible to feathery bits floating around in the EDTA. There should be no mineralized chunks remaining. If there are, leave the samples to further demineralize for an extra couple of days.

Note: If the room temperature is > 30 °C or fluctuates greatly, demineralization can be performed at 4 °C or room temperature in an incubator.

Note: If not immediately proceeding to Day 1 of SP3, samples can be stored in the freezer (- 20 °C). Thaw samples completely after freezing before continuing.



IV SECTION: SP3 PROTOCOL DAY1

Note: Start this section only when your samples are adequately demineralized.

All preparations and pretreatment steps should take place in the dedicated extraction hoods.

Before start to work:

- Wipe down all surfaces
- Check the dates on the reagents and the shelf life before you start
- **Aliquot out in 5mL Eppendorf tubes** (roughly calculate the amount you need!):
 - 6 M GuHCl (stored at 4°C , 200 µl per sample needed)
 - 100 mM ammonium bicarbonate (stored at 4°C, 75 µl per sample needed)
 - Molecular grade H₂O (1 ml in sum needed, Open bottles are stored at 4°C and new and sealed bottles are inside the reagents cabinet)
- **Aliquot out in 15mL or 50mL tubes**
 - Ethanol 100% (stored at 4°C, 350 µl per sample needed)
 - Ethanol 80% (stored at 4°C, 600 µl per sample needed)
- **Check that enough of the following are available:**
 - Trypsin stock solution (0.2 µg/µl, stored at -20 °C in the reagents drawer, 2 µl per sample needed → Check the date! Trypsin is good for a month stored at - 20°C once resuspended. If there is none you will need to resuspend it. Please check the Reagents Instructions Section!
 - Working bead solution (20 µg/µl, stored at 4°C, check date! Bead solution shall be not older than one month, 20 µl per sample needed)
 - TCEP (stored at 4°C, 28.66 mg pre-weighed)
 - CAA (black tubes stored at 4°C , 9.35 mg pre-weighed)
 - Stage tips (required from Day 2; but you should not start Day 1 without checking)
 - Clean magnetic racks

3. Demineralization continued

- Turn the Thermoshaker to 99°C in preparation for the heating step (Program S1 at the Thermomixer).
- Remove all samples from the rotator and centrifuge at max speed ~20,000x g (RCF) for 10 min to separate supernatant from the pellet.



- During centrifugation, label a new set of microcentrifuge tubes as follows:
 - Top: DA 206
EDTA Sup
 - Side: same as previous set of tubes
- Prepare the 100 mM CAA/100mM TCEP solution.
 - Add 1 ml molecular grade water to the TCEP preweighed tube.
 - Vortex shortly
 - Transfer the TCEP solution to a black CAA preweighed tube
 - Vortex 1 min
- Remove 400 µl of the original samples' supernatant and transfer to the tubes labelled "EDTA Sup." Store at -20 °C for backup (place in a labelled sample box or bag). You should have 100 µl and the pellet remaining in the original sample tube.

4. Denaturing, Reduction, Alkylation and Lysis

- Add 200 µl of the 6 M GuHCl to each tube and resuspend to mix.

Note: You should see most, if not all, of the pellet disappear. If you have humics, such as dirt or other non-mineralized materials in your sample, these may still be in the pellet and they will be removed in later rinse steps.

- Briefly centrifuge the samples to get the liquid to the bottom of the tube ("SHORT").
- Add 30 µl of the prepared 100 mM CAA/100 mM TCEP solution to each tube.¹
- Vortex each sample quickly (~ 2 seconds) and place the samples in the Thermoshaker for 10 min, 0 rpm at 99 °C (Program S1 at the Thermomixer).
- While waiting for your samples on the Thermoshaker, label a new set of microcentrifuge tubes with the DA code and "SP3 Sup" on the top of the tube and the full details on the side.
- After 10 min, remove your tubes and give your samples some time (approximately 5 min) to cool.
- Turn the ThermoMixer down to 24°C. (Program S2 at the Thermomixer; If necessary: use the prepared microcentrifuge tubes of ice (stored in -20 freezer) to speed up the cooling process. Be sure to return the tubes of ice to the freezer once the required temperature has been reached).
- Briefly centrifuge the samples to get the liquid to the bottom of the tube ("SHORT")

¹ Final working concentration should be 10 mM CAA/ 10 mM TCEP. For different starting volumes, adjust.



5. Protein Purification

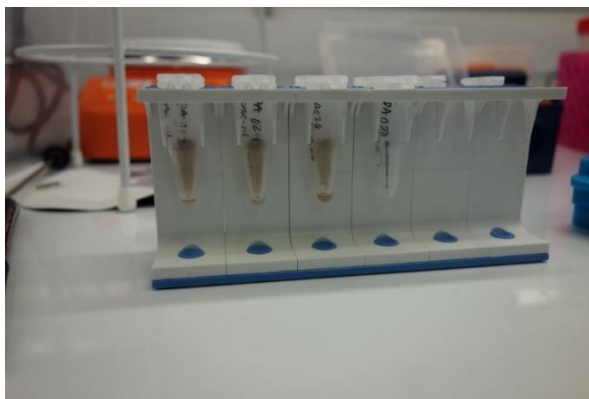
- Add 20 μl of the working bead solution (20 $\mu\text{g}/\mu\text{l}$) to each tube to begin SP3 clean-up and mix by gentle pipetting (resuspension).²

Note: Ensure complete homogenization of the SP3 beads in the solution. Improper mixing will reduce the protein recovery. Once the beads are added, the sample cannot be stored frozen until the beads are removed.

- Add 350 μl 100% Ethanol to the solution in the same volume as your sample and briefly shake the tube by hand to mix.³

Note: Avoid excessive shaking and do not flick or invert the tube. Both can cause protein loss due to beads sticking to the upper areas of the tube. Complete homogenization is not essential as it will be achieved during the incubation on the Thermoshaker.

- Incubate the tubes on the Thermoshaker for 5 min at 1000 rpm at 24°C (Program S2 at the Thermoshaker).
- Place the tubes on the magnetic rack and wait for the beads to migrate to the wall (approx. 1 min).



Note: Samples containing protein will have a fan shaped appearance against the wall of the tube (see photo below). Blanks and samples with very low protein will appear as a thin line of beads along the wall of the tube.

- Remove the sample/Ethanol solution, without disturbing the beads, and place in the tubes labelled “SP3 Sup”. Store at -20 °C as a backup.

² Final bead concentration at this point is 1.14 $\mu\text{g}/\mu\text{l}$. Recommendations say that during binding (after the addition of Ethanol) the ratio of beads to protein should be 1:10, but no lower than 0.5 $\mu\text{g}/\mu\text{l}$. Adjust the amount of bead solution used if you use a different volume or know you have an extremely high protein concentration.

³ Volume of 100% Ethanol should be the same as the volume of solution in the tube.



Note: Calculus samples can be extremely “dirty” and the beads may become sticky. If the beads stick to the walls gently push them with a pipette tip back into the liquid so they can fully migrate to the tube wall which is against the magnetic rack. Each rinse should reduce this “stickiness” and the beads should easily migrate to the wall by the end. There is no harm in performing an extra rinse if needed!

- Remove the tubes from the magnetic rack and add 200 µl 80% Ethanol and resuspend the beads in the solution to rinse.

Note: Aggressive or excessive pipetting during washing can result in protein loss.

- Put the tubes back on the magnetic rack, wait for the beads to migrate to the magnetic wall.
- Remove and discard the supernatant without disturbing the beads.
- Repeat 80% Ethanol wash two additional times for a total of three washes. ☐ ☐ ☐
- After the final rinse, be sure to remove as much of the Ethanol supernatant as possible without disturbing the beads.

Note: You should not leave more than 5 µl in the tube.

6. Digestion

- Remove the tubes from the magnetic rack and add 75 µl of 100 mM ammonium bicarbonate solution to each sample.

Note: Use the tip of a pipette to push any beads down into the digestion solution, but do not actually pipette/resuspend the solution at this point. Also do not shake, flick, or invert the tube.

- Remove the trypsin solution from the -20 °C freezer for thawing.

Note: Trypsin should be removed just prior to use (leaving enough time to thaw) and then immediately returned to the freezer.

- Incubate the tube at 37 °C in the Thermoshaker for 10 min at 750 rpm (Program S3 at the Thermomixer) or sonicate your sample for 30 sec at RT to disaggregate the beads
- Add 1 µl of the trypsin solution to each tube.⁴

⁴ This will end up with 0.2 µg of trypsin to each sample. To use a different amount, adjust the amount of trypsin accordingly.



- Mix your sample again through up and down pipetting to ensure an equal distribution of the beads and the trypsin.
- Now incubate your samples for 18 hrs at 750 rpm in the Thermoshaker (Program S4 at the Thermomixer).
- Put the tube with the trypsin solution back into the -20 freezer.



V SECTION: SP3 PROTOCOL DAY 2

Before start to work:

- Wipe down all surfaces
- Check the dates on the reagents and the shelf life before you start
- Aliquot out into 1.5ml Eppendorf tubes
- 5% TFA (20 µl per sample)
- **Aliquot out in 50mL Eppendorf tubes**
 - o Methanol (150 µl per sample needed)
 - o 60% ACN, 0.1% TFA (150 µl per sample needed)
 - o 3% ACN, 0.1% TFA (600 µl per sample needed)
- **Check that enough of the following are available**
 - o Stage tips
 - o Clean stage tip holders

7. Acidification:

- Remove the samples from the Thermoshaker and centrifuge them "SHORT".
- Place the tubes on the magnetic rack and wait until all the beads migrated to the wall.
- As you wait, label a new set of tubes with just the DA number on the top (these will not be retained after use).
- Transfer the entire supernatants to the labelled tubes and discard the tubes with the remaining beads.
- Acidify the samples by adding 10 µl of 5% TFA to bring the pH below 2.5.
- Quickly vortex the sample and test the pH using 1-4 µl of sample on the pH strips. If 10 µl does not lower the pH enough, add an additional 5 µl of 5% TFA until the desired pH is reached (< 2,5). Record in your lab notebook the total volume of 5% TFA added to each sample and the resulting pH to keep track of any anomalies.

Note: It is essential to acidify the samples in order to stop the tryptic digestion and to enable the peptides to adhere to the stage tip C18 material.



8. Peptide Purification

- Prepare 2.0 ml Eppendorf tubes by placing the stage tip holders into 2.0 ml tubes to allow the StageTips to be inserted.
- Label the StageTips with the DA sample number 3 times.

Note: It is necessary to clean the StageTips before loading the samples. Since the waste from cleaning the StageTips will be discarded prior to peptide collection, it is not necessary to label this set of 2.0 ml tubes with your sample IDs.

Note: Be careful when handling the stage tips. Discard any stage tips if the filters are not properly situated in the tips or are disrupted during handling.

- Prepare a second set of 2.0 ml tubes with the “L+W” and the sample number on the top, and fully note with sample number, date, and “Load/Wash” on the side.
- Add 150 µl of Methanol (MeOH) to the StageTip. Centrifuge at 2000x g (RCF) for 1 minute.
- Add 150 µl 60% ACN, 0.1% TFA to the tips and centrifuge at 2000x g (RCF) for 1 min.
- To equilibrate the StageTips, add 150 µl of 3% ACN, 0.1% TFA and centrifuge at 2000x g (RCF) for 1 min. Make sure all liquid has passed through the tip and into the tube.
- Repeat the above step (add 150 µl of 3% ACN, 0.1% TFA and centrifuge at 2000x g (RCF) for 1 min).

Note: Check your StageTips after every centrifugation step to see if all the liquid went through and that the filter is still properly situated in the StageTip. If the filter has become dislodged, discard the StageTip and prepare a new one.

- Carefully transfer the StageTip in its holder in the holder to the 2 ml tube labelled “Load/Wash”.
- Discard the waste accumulated in the tubes into the halogenic waste container and discard the empty, unlabeled tubes.
- Centrifuge the tubes containing your acidified samples for a short spin.

Note: If samples contain particles or undigested parts of the pellet at this point, you can spin longer or at higher speeds in order to avoid transferring accumulated particles to the StageTip as these could clog it.



- Place the acidified sample tubes back on the magnetic rack to make sure no remaining beads are transferred to the StageTip.

Note: Avoid transferring any beads or any particulates to the StageTip as these could clog it.

- Transfer the sample to the StageTip and centrifuge for 3 min at 2000x g (RCF). Check now to see that the entire sample has passed through each StageTip! If the sample does not pass through, continue centrifuging until the samples has passed the tip.
- Add 150 µl of 3% ACN, 0.1% TFA to the Stage Tip and centrifuge at 2000x g (RCF) for 2 min. Again, check every single stage tip that the liquid passed through the stage tip. If the liquid did not pass, continue centrifugation until it has completed.
- Repeat the above step (add 150 µL of 3% ACN, 0.1% TFA to the Stage Tip and centrifuge at 2000x g (RCF) for 2 minutes).
- Store the “Load/Wash” tubes in a labelled bag or box in the -20 freezer.
- Store the stage tips containing the peptides in the -20 freezer until they are taken to Zürich for MS/MS analysis.

Note: If eluting: Transfer StageTip with the holder to a new tube and add 60 % ACN, 0.1 % TFA to the StageTip and centrifuge at 2000x g at room temperature for 2 min.